

IM1 Book 1 Selected Answers

IM1 Dream Team

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1. (a) $A_{ABCD} = 25$, $A_{BCEF} = 9$
(b) –
(c) –
(d) $A = 34$
(e) $l = \sqrt{34}$
(f) –
2. $l = 4\sqrt{5}$
3. –
4. $AB = \sqrt{41}$
5. $l = 5\sqrt{2}$
6. $l = \sqrt{5}$, No
7. $d = 10\sqrt{2}$
8. 12
9. $(12, 2), (2, 2)$
10. No
11. (a)

$$SR = PQ = 13$$

$$RQ = PS = 17$$

$$m_{SR} = m_{PQ} = \frac{-5}{12}$$

$$m_{RQ} = m_{PS} = \frac{15}{8}$$

- (b) It is a parallelogram but not a rectangle.

12. $(-6, 3), (10, 3)$

13. $AB = BC = \sqrt{10}$
14. (a) $(5, 0)$ – Answers may vary.
 (b) $(5, 1)$ – Answers may vary.
 (c) $x = 5$
 (d) –
15. (a) $AP = BP = 2\sqrt{5}$
 (b) $(2, 2), (3, 5), (4, 8)$
 (c) No
 (d) $y = 3(x - 1) - 1$
16. (a) $(2, 5), (3, 4)$ – Answers may vary.
 (b) $(1, 4), (2, 3)$ – Answers may vary.
17. $(1, 5)$ – Answers may vary.
 There are infinite options.
18. (a) Same size, different shape, incomparable orientations – the pieces aren't equal.
 (b) Same size, same shape, different orientations – the pieces are equal
- Definition: Two figures are congruent if they have the same size and shape.
19. $y = \frac{1}{3}(x - 2) + 2$
20. $(0, 5), (3, 4), (4, 3), (5, 0), (4, -3), (3, -4), (0, -5), (-3, -4), (-4, -3), (-5, 0), (-4, 3), (-3, 4)$
21. $(2, 3), (2, -3)$
22. (a) Answers may vary: $(0, 0), (2, 5)$.
 (b) No
23. Answers may vary: $(0, 0), (4, 1), (8, 0), (4, -1)$
24. $(0, 5 + 4\sqrt{2}), (0, 5 - 4\sqrt{2})$
25. 6
26. Leg 1: $24 - 12\sqrt{2}$
 Leg 2: $24 - 12\sqrt{2}$
 Hypotenuse: $24\sqrt{2} - 24$
27. Yes
28. (a) If point P is equidistant from the coordinate axes, then point P is on the line $y = x$.

- (b) No
 - (c) Answers may vary. If quadrilateral $ABCD$ is a square, then quadrilateral $ABCD$ is a rectangle.
 - (d) Answers may vary. If quadrilateral $ABCD$ has four right angles, then quadrilateral $ABCD$ is a rectangle.
29. They have the same side lengths.
30. (a) $\triangle ABC$ must be the same size and shape as $\triangle DEF$.
If two triangles have the same three side lengths, then the two triangles are congruent.
- (b) $\overline{KL}$
- (c) Typo: The question should say “What angle of $\triangle ABC$ corresponds to $\angle LKM$?”
Answer: $\angle BAC$
- (d) $\overline{GI} \cong \overline{GI}$, so there are three pairs of congruent segments.
- (e) They are congruent.
31. –
32. $\angle ABC$
33. $\overline{AB}$
34. (a) $PNMRQ$
(b) $\angle Q$
35. They must add to 90° .
36. It must be 90° .
37. $d = \sqrt{(x-2)^2 + (y-4)^2}$
38. (a) –
(b) –
(c) –
(d) –

$$m_{QR} = \frac{-a}{b}$$

$$m_{QP} = \frac{b}{a}$$

39. –

40. –

41. –

42. –

43. –

44.

$$\triangle LAG \cong \triangle LEB$$

$$\triangle LEG \cong \triangle LAB$$

$$\triangle AGB \cong \triangle EBG$$

$$\triangle AGE \cong \triangle EBA$$

45. –

46. (a) $y - 2 = \frac{1}{2}(x + 1)$

(b) $y = \frac{4}{5}(x - 2)$

(c) $y = -\frac{5}{4}(x - 2)$

47. (a) –

(b) Q; C.P.C.T.C.

48. $n = \frac{49}{4}$

49. (a) $\overline{AC} = \overline{AB} + \overline{BC}$; segment addition

(b) –

50. Yes

51. No

52. C.P.C.T.C.

53. (a) $\angle ABD + \angle DBC = \angle ABC$; angle addition

(b) –

54. 18 seconds

55. The slope is the speed of the plane's descent.

The y -intercept is the plane's location when initially measured.

The x -intercept is the plane's location when it lands.

56. $m = -1$

57. –

58. –

59. (a) $\sqrt{(x + 1)^2 + (y - 5)^2}$

- (b) $\sqrt{(x-5)^2 + (y-2)^2}$
 - (c) $\sqrt{(x+1)^2 + (y-5)^2} = \sqrt{(x-5)^2 + (y-2)^2}$
 - (d) $y = 2x - \frac{1}{2}$
 - (e) $(2, \frac{7}{2})$
 - (f) $m_{AB} = -\frac{1}{2}$
 - (g) $-$
60. All points (x, y) that are equidistant to $(3, 5)$ and $(7, -1)$.
61. (a) $y - 5 = \frac{5}{4}(x - \frac{1}{2})$
 (b) $(\frac{9}{2}, 10)$
 (c) $\overline{PA} = \overline{PB}$
62. 560 ft
63. $(-\frac{7}{2}, -\frac{1}{2})$
64. (a) No
 (b) If two triangles have two pairs of congruent sides and the angles formed by the two pairs of sides are congruent, then the two triangles are congruent.
 (c) Yes
 (d) No
 (e) If the angle is acute, there can be two non-congruent triangles that have S.S.A. congruencies
65. $-$